RAMCO INSTITUTE OF TECHNOLOGY
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Department of Computer Science and Engineering
Academic Year 2021 – 2022 (Even Semester)

Degree, Semester & Branch: B.E.VI, Computer Science and Engineering
Course Code & Title: CS8075 Data Warehousing and Data Mining
Name of the Faculty member (s): Mrs.M.Swarna Sudha

Innovative Practice Description

Unit 2-Topic: Data Visualization

Course Outcome: CO2

Topic Learning Outcome: 2.3

Activity Chosen: Learning by Doing

Justification:
Data visualization provides a good, organized pictorial representation of the data which makes it easier to understand, observe, analyze . Data visualization is an important aspect of all AI and machine learning applications. Students will get in-depth knowledge of the particular topics by doing.. The prime objective of Learning by doing is to enhance the problem solving and creative thinking of the students

Time Allotted for the Activity: 20 minutes

Details of the Implementation:

- Dataset were given to the students prior to the start of the event for preparation purposes.
- After completion of the unit-II preprocessing and statically description of data , the students involved in this activity.
- Each and every student upload file Google colab in canvas
- Students were asked to interact and share their ideas in Data visualization Finally at the end of the session concepts were explained and discussed.

- **CO – PO / PSO mapping:**

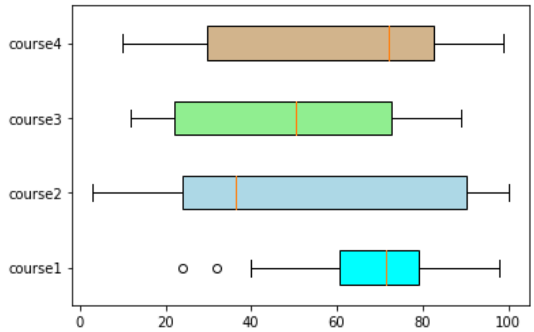
CO	PO1	PO2	PO3	PO4	PO9	PO10	PO12	PSO3
CO2	3	1	1	2	2	2	2	2

(1 – Low 2 – Moderate 3 – High)

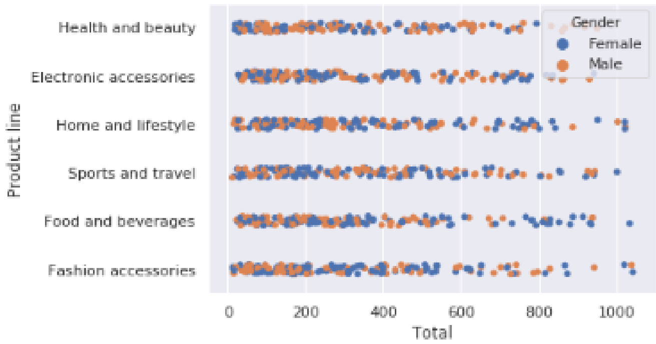
PO / PSO mapped:

Innovative practice	**PO1**	**PO2**	**PO3**	**PO4**
	3	**1**	**1**	**2**
Justification for correlation	Data visualization is basic concept necessary to do analysis	Data visualization is used to analysis the engineering problems	To analysis the complex problems data visualization is used.	To provide in-depth knowledge of problem solution with visualization
	PO9	**PO10**	**PO12**	**PSO3**
	2	**2**	**2**	**2**
	Doing effectively as an individual, by understand the concept	Communicate effectively on complex engineering activities by visualization the problem solution	To provide to have the preparation and ability to engage in independent by doing the concept	To providing solutions to real world problems in Industry by visualizing the solution

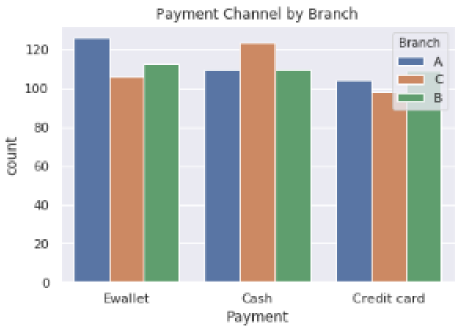
Screenshot of the Google Colab practice:



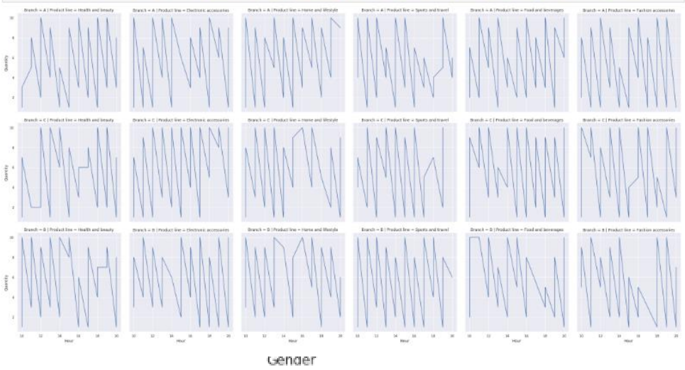
In [28]:
```
sns.stripplot(y = 'Product line', x = 'Total', hue = 'Gender', data=sales )
```
Out[28]: <matplotlib.axes._subplots.AxesSubplot at 0x7f1409bfc7f0>



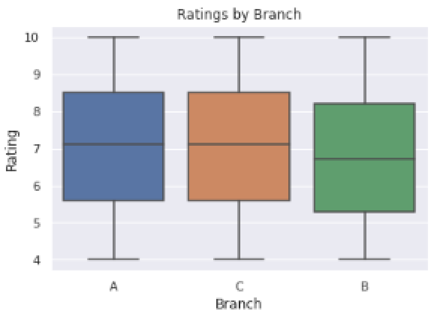
In [33]:
```
sns.countplot(x="Payment", hue = "Branch", data =sales).set_title("Payment Channel by Branch")
```
Out[33]: Text(0.5, 1.0, 'Payment Channel by Branch')



In [31]:
```
productCount = sns.relplot(x="Hour", y = 'Quantity', col= 'Product line' , row= 'Branch', estimator = None, kind="line", data =sales)
```



In [19]:
```
sns.boxplot(x="Branch", y = "Rating" ,data =sales).set_title("Ratings by Branch")
```
Out[19]: Text(0.5, 1.0, 'Ratings by Branch')



Branch B has the lowest rating among all the branches

Reflective Critique:

- The problem-solving capabilities of students will be improved.
- It is helpful for the students to understand the concept of data visualization in depth

Benefit of the practice:

- Students can able to attend the question even in the questions are in indirect form.
- Students can able to explain the concepts without any confusion.
- Students understood the concept which was reflected from their answers in unit II quiz for the of Data visualization as shown in the fig 2

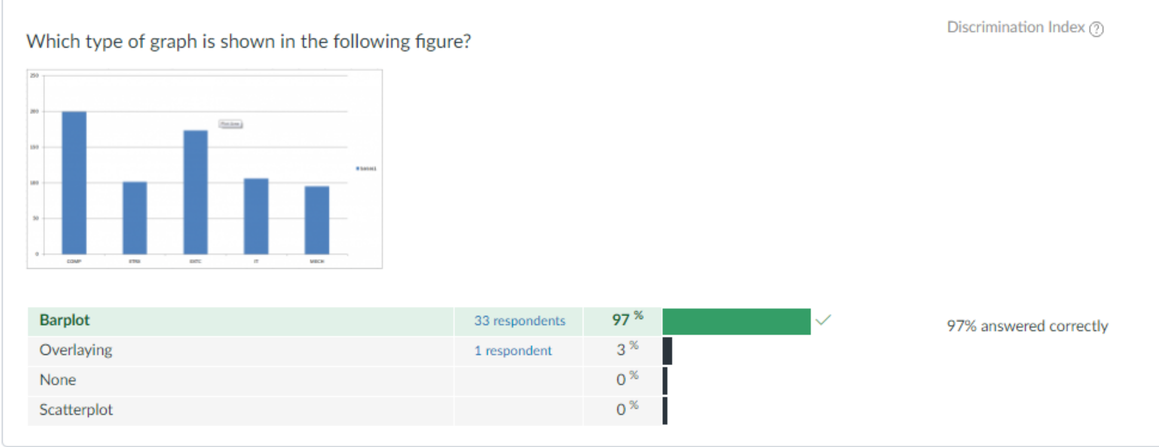


Fig 2 In Unit II quiz 97 % of the students have answer correctly

- ***Challenges faced in implementation:***
- Students learned a few things individually that **leads to self-learning**. They also actively involved in this session rather than it being a one-way communication.
- I felt slow learners were not comfortable in solving because of their individual (slow) learning background. However, they understood the concepts through the interaction within the team.

References:

1. https://opentextbc.ca/teachinginadigitalage/chapter/4-4-models-for-teaching-by-doing/ - 3.6.3.2 Problem-based learning
2. https://www.ritrjpm.ac.in/images/computer-science/Learning-by-doing(TOC).pdf

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